Coding & Solution

Date	3 July 2026
Team ID	xxxxxx
Project Name	Rising Waters - Flood Prediction System
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/AlRihabChandhiniMohammed/rising-waters-flood-prediction
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn

Key Features Implemented	• Data preprocessing (handling missing values, categorical values, feature scaling) • Decision Tree, Random Forest, KNN and XGBoost model implementation • Model evaluation and best model selection • Flood prediction using trained model • Flask web application with user-friendly interface • Flood prediction result pages (Flood / No Flood)
Pending / Incomplete Features	• IBM Cloud deployment (optional) • Real-time weather API integration • User authentication and prediction history
Setup / Run Instructions	1. Clone the GitHub repository. 2. Install dependencies using pip install -r requirements.txt. 3. Run python app.py. 4. Open http://127.0.0.1:5000 in a web browser. 5. Enter weather parameters and click **Predict** to view the result.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The project implements a complete machine learning pipeline for flood prediction, including data preprocessing, model training, evaluation, and deployment using Flask. The repository follows a structured folder organization with separate directories for datasets, models, templates, static files, and source code. The application provides an intuitive interface for users to enter weather parameters and receive flood prediction results. The codebase is modular, maintainable, and suitable for future enhancements such as cloud deployment and real-time weather data integration.